AI-Based Crop Recommendation Framework for Farmers

1 Kaif Islam , 2 Vishal Kumar Chaurasia, 3 Rohit Kumar, 4 Abhishek Kumar, 5 Raman Goyal

1,2,3,4,5 Department of Computer Science Engineering, Lovely Professional University, Jalandhar (Punjab), India

1 kaifislam48@gmail.com, 2 vkc140@gmail.com, 3 rksanda49@gmail.com 4 rajabhishek8070@gmail.com
5 ramangoyal87021@gmail.com

***Abstract*:-** Agriculture is really important for feeding the world. Picking the right crop for the local soil and environment makes a big difference in the harvest. This study talks about a system that suggests the best crop by using smart technology. It considers factors like soil nutrients—Nitrogen, Phosphorus, Potassium—as well as temperature, humidity, rainfall, and soil pH. The system checks out machine learning models to find the best one. It looks at Logistic Regression and Support Vector Machine and Random Forest. The system wants to know which of these machine learning models works best. It also uses Boosting techniques to help the machine learning models. The results show that some models, Random Forest work very well and are accurate. The proposed system is useful can handle amounts of data and can help with real-world farming decisions. The crop recommendation system uses soil nutrients and environmental factors to make suggestions. It helps farmers choose the crop for their soil and environment. The system is based on AI. Uses machine learning to make predictions. It is practical. Can be used in real-world farming.

Keywords:- Machine Learning, Crop Recommendation, Precision Agriculture, Soil Analysis, Classification Models, Smart Farming

1. INTRODUCTION

Agriculture supports the economy, especially in the developing countries where many people depend on farming for their income and livelihood. It plays a role in food production, job creation, economic stability and rural development. As the global population keeps growing the need for food production is increasing rapidly putting pressure on agricultural systems to boost productivity and efficiency. One important factor that affects how well crops grow is picking the right plant for the soil and environment. Nutrients in the soil, like nitrogen, phosphorus, and potassium, are necessary for plants to grow and produce a good harvest. Things like temperature, humidity, and rainfall also have a big impact on which crops do well. The soil's pH level can change how nutrients are available and affect the microbes living in the soil. Since this conditions can change a lot of depend on the location and time of year, choosing the best crop can be quite challenging. Traditionally farmers rely on their experience, intuition and conventional agricultural practices to decide which crop to cultivate. While such knowledge is valuable it often lacks precision. May not adapt well to changing environmental conditions, soil degradation and climate variability. Because of this, farmers might choose crops that don't fit their land well, which can lower their harvest, waste fertilizers and water, and increase financial risks. Recently, progress in Machine Learning has created new chances to change farming into a smarter, data-based process. These machine learning tool can study large amounts of data, spot patterns, and make reliable predictions using different factors. These capabilities make ML particularly suitable for solving agricultural problems like crop recommendation, yield prediction, disease detection and resource optimization. Using AI in farming helps farmers make choices based on data. Machine learning can create systems that suggest best crops for specific soil and environmental conditions. These systems have the potential to boost crop yields, cut down on wasted resources, and support more sustainable farming. This research works on building an AI crop recommendation system that uses several machine learning algorithms to predict suitable crops. It looks at important factors like soil nutrients(N, P, K), temperature, humidity, rainfall, and pH. By comparing different models, the study finds which algorithms works best for this. The goal is to offer reliable, real-time advice that farmers can actually use in their decisions. The system checks its performance across the various crop types using measure like accuracy, precision, recall, and F1-score. Ultimately, this research tries to connect farming with newer tech by offering a solution that can grow and adapt. Farmers using this system might choose crops better suited to their land, improving productivity and saving resources. AI in agriculture can also be contribute to sustainable practices and food security. Since farming is a key part of many economies, applying the machine learning techniques could help improve how farming is done by making data-driven choices that lead to better harvests and less waste.

2. PROBLEM STATEMENTS

Agriculture productivity really depends on choosing the crop for the soil and the environment. Farmers usually make decisions based on what they know their experience or what other people tell them.. This doesn't always work because soil has different levels of nutrients like nitrogen, phosphorus, and potassium. Weather also matters, including temperature, humidity, and rainfall. We have to consider soil pH as well. If farmers don't pick the right crops, they won't get good yields from the land and might waste resources and lose money. Some systems try to help farmers, but they aren't very effective. They often don't use enough information to make decisions and can't handle many factors at once. Their advice on what to plant isn't very accurate. Many of these systems don't use machine learning and cannot process large amounts of the data quickly. So, there's a need for a system that helps farmers decide on crops based on various soil and environmental factor. The AI-Based Crop Recommendation System offers a solution by using machine learning algorithms to analyze these factors and suggest the best crops. These can improve agricultural productivity, make farming more sustainable, and help farmers make better decisions. In short, this system uses machine learning techniques to recommend the right crops for each farmer's land.

3. LITERATURE RIVIEW

Machine learning is increasingly being used in farming to help growers produce food more efficiently and manage resources better. People are exploring how to classify and predict things like which crops to plant, how much yield to expect, and soil health. Techniques like Decision Trees and

Random Forests are quite popular because they're easy to understand and handle relationships well. Decision Trees work like map that show how different factors connect, which make them useful for the farming decisions. The Random Forest combine many Decision Trees to make predictions more accurate and has been widely applied to forecast crop yields with good results. Besides these tree-based approaches, others use networks—computer systems that can learns from large amounts of data. These are great at spotting patterns in images or big data but need lots of data and computing power, which can be a challenge for farmers with limited resources. The Support Vector Machines are another option, handling complex data by separating different types of crops, but they require careful setup and take time to tune. Despite their usefulness, there are some issues with these machine learning methods. Many studies focus on just one model, making comparison difficult. Some datasets aren't balanced, meaning certain crops get prioritized, which can lead to unfair or unreliable results. Also, many existing systems aren't designed for practical use on farms and don't measure their performance fully. To tackle all this problems, the study examines differents machine learning model using the balanced dataset covering 22 crop types. It also uses various performance metrics to judges how well each model works and aims to design a system that can be applied directly in farming. The goal is to build a reliable tool that predicts crop yields accurately and supports farmers in making better decisions. This study employs linear kernel methods, instance-based approaches, and ensemble techniques to reach that goal. By using a well-rounded dataset and multiple ways to measure performance, it hopes to create a fair system farmers can depend on to boost their productivity and resource use. The system will also be designed to deliver timely information so farmers can act quickly. Applying machine learning in farming holds promise, and this study contributes by developing a practical system for real-world use. It focuses on fairness and reliability through balanced data and thorough evaluation. The ultimate aim is to help the farmers grow food more effectively and use resources smarter, moving both farming and technology forward together.

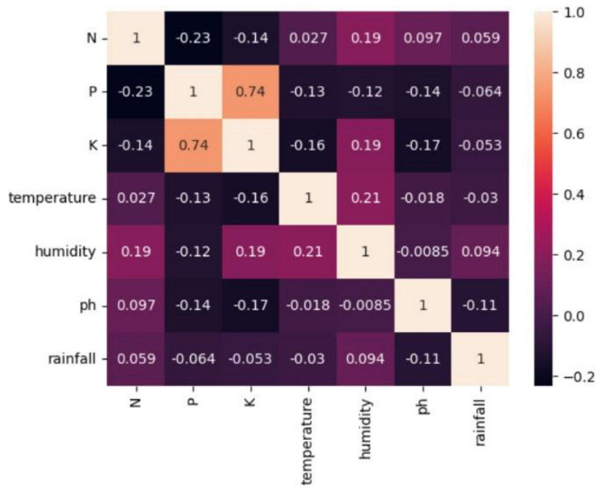

Figure1:- Overview of Datasets

Feature Description

Feature	Description
Nitrogen (N)	Soil nutrient level
Phosphorus (P)	Soil nutrient level
Potassium (K)	Soil nutrient level
Temperature	Environmental temperature (°C)
Humidity	Relative humidity (%)
pH	Soil pH value
Rainfall	Rainfall (mm)

Table1:- Feature Description

4. DATASET DESCRIPTION

The dataset consists of 2200 samples with 7 input features and 1 target variable.
Input Features:-

- Soil Nutrients: N, P, K
- Environmental Factors: temperature, humidity, rainfall
- Soil Property: pH

Target Variable:-

- Crop type (22 categories)

The dataset is:-

- Balanced (100 samples per crop)
- Clean (no missing or duplicate values)

This ensures unbiased training and reliable evaluation.

4. METHODOLOGY

4.1 Data Preprocessing:-
Data preprocessing is a crucial step to ensure data quality, consistency, and model performance.

- Removal of inconsistencies
- Label encoding of target variable
- Feature scaling using StandardScaler

$$X' = X-\mu / \sigma$$

Where:
X = original feature value
μ = mean of the feature
σ = standard deviation

4.2 Train-Test Split:-
The dataset is divided into:

- 80% training data
- 20% testing data
- Ensures proper model generalization

Mathematically:

$$D = D_{train} \cup D_{test}$$

$$|D_{train}| = 0.8N, \ |D_{test}| = 0.2N$$

Purpose:

- Train model on known data
- Evaluate on unseen data

4.3 Machine Learning Techniques Used:-

The system evaluates multiple classification algorithms to identify the best-performing model.

- Linear Model: Logistic Regression
- Probabilistic Model: Gaussian Naive Bayes
- Kernel-Based Model: Support Vector Machine
- Instance-Based: K-Nearest Neighbors
- Tree-Based: Decision Tree, Extra Tree
- Ensemble Methods:
  Random Forest
  Bagging Classifier
  Gradient Boosting
  AdaBoost

4.4 Evaluation Metrics:-

Models are evaluated using:

- Accuracy Score
  Accuracy = TP+TN / TP+TN+FP+FN
- Precision Score
  Precision = TP / TP+FP
- Recall Score
  Recall = TP / TP+FN
- F1 Score
  F1=2× Precision · Recall / Precision + Recall

Macro-averaging is used due to multi-class classification.

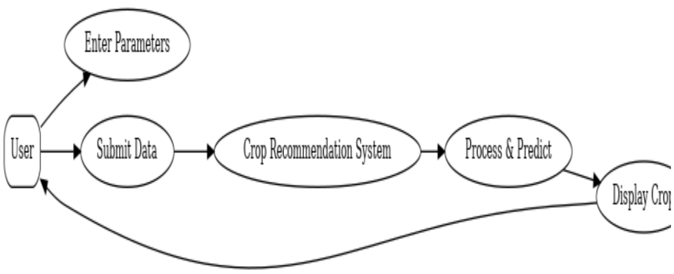

Figure2:- Use Case Diagram

The figure2 represents the interaction between the user and the AI-Based Crop Recommendation System, illustrating how input the data is processed to generate crop recommendations.

Components of the Diagram:-

1. User

- The primary actor in the system
- Represents farmers or end-users

Role:

- Provides input parameters
- Receives crop recommendation

2. Enter Parameters

This step involves inputting agricultural data such as:

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- Temperature
- Humidity
- pH
- Rainfall

These inputs define the current soil and environmental conditions.

3. Submit Data

- The user submits the entered parameters to the system
- Acts as a trigger for the prediction process

4. Crop Recommendation System

- Core component of the system
- Contains the trained machine learning model

Function:

- Receives input data
- Sends it for processing

5. Process & Predict

This is the main computational stage, where:

- Input data is preprocessed (scaling, normalization)
- Passed to the trained model (e.g., Random Forest)
- Prediction is generated

Internally includes:

- Feature transformation
- Model inference

6. Display Crop

- Final output stage
- Displays the recommended crop to the user

Example:

- Output: Rice

4.5 Evaluations Metrics

Random Forest Classifier Performance

Accuracy : 0.9932
Precision : 0.9926
Recall : 0.9933
F1 Score : 0.9926

5. RESULTS AND MODEL COMPARISON

a) Logistic Regression

A linear model used for classification.

$P(y=1|x) = 1 / 1+e-(wTx+b)$

b) Gaussian Naive Bayes
Based on Bayes' Theorem:

$P(C|X) = P(X|C) \cdot P(C) / P(X)$

Assumes:

- Features are conditionally independent
- Data follows Gaussian distribution

c) Support Vector Machine (SVM)
Finds optimal hyperplane:

$W \cdot x + b = 0$

d) K-Nearest Neighbors (KNN)
Distance-based classification:

$d(xi ,xj)=\sqrt{\sum(xi - xj)2}$

Prediction based on nearest neighbors.

e) Decision Tree
Splits data using impurity measures:

$G = 1 - \sum pi2$

f) Ensemble Methods
Random Forest

- Combination of multiple decision trees
- Final prediction:

$y\hat{} = \text{mode}(T1,T2,...,Tn)$

Bagging

- Reduces variance using bootstrap sampling

Gradient Boosting

- Sequentially improves errors

AdaBoost

- Assigns weights to misclassified samples

Model	Accuracy
Logistic Regression	0.95
Gaussian Naive Bayes	0.99
Support Vector Machine	0.97
K-Nearest Neighbors	0.98
Decision Tree	0.99
Random Forest	0.99 – 1.00 (Best)
Bagging Classifier	0.99
Gradient Boosting	0.99
AdaBoost	0.97

Table2:- Model Accuracy Comparison

6. DISCUSSION

The test results comparing differents machine learning techniques show that using a group of models together tends to suggest crops more accurately than relying on a single model. Among all models tested Random Forest did the best in all areas, like accuracy, precision, recall and F1-score. One key thing we found out is that ensemble models are better at generalizing than models. This is mainly because ensemble techniques combine predictions from the base models, which reduces model variance and makes it more robust. Random Forest makes many decision trees using part of data and features and combines their predictions using a majority vote. This makes predictions more stable. Reduces the chance of overfitting. Random Forests great performance is also because it can handle relationships between input features. Agricultural data like the one we used has dependencies between soil nutrients (N, P, K) weather conditions (temperature, humidity, rainfall) and soil pH. Linear models like Logistic Regression may struggle with these interactions. In contrast tree-based ensemble methods can handle these complexities well leading to predictions. Another important factor is the nature of our dataset. It is balanced with a number of samples for each crop category. This eliminates class imbalance issues ensuring the models does not become biased toward classes. As a result performance metrics like precision, recall and F1-score are consistents across all classes, which's particularly important in multi-class classification problems. In comparison probabilistic models like Gaussian Bayes also achieved high accuracy. However their assumption of features independence may not always be true in world agricultural scenarios. Similarly Support Vector Machines demonstrated performance but required careful parameter tuning. They were also computationally more intensive making them less suitable for the real-time applications. Instance-based methods like K-Nearest Neighbors performed well but are sensitive to feature scaling. They are computationally expensive during prediction for large datasets. Decision Trees, while interpretable are prone to overfitting when used individually. This explains their lower generalization compared to ensemble methods. The use of evaluation metrics provides a comprehensive understanding of model performance. Accuracy indicates correctness. Precision and recall offer insights into class- performance. The F1-score provides a balance between the two. The use of macro-averaging ensures that all crop classes are treated equally. This reinforces the fairness and reliability of the evaluation. Overall the results confirm that ensemble learning techniques, Random Forest are highly effective for crop recommendation tasks. Random Forests ability to handle dimensional data, model complex relationships and provide robust predictions makes it well-suited for real-world agricultural applications. The findings of the study highlight the importance of using advanced machine learning techniques to enhance decision-making in precision agriculture. The results show Random Forest and ensemble methods are tools for crop recommendation. Random Forest is particularly effective in handling agricultural data. Ensemble methods like the Random Forest can improve decision-making, in precision agriculture.

7. SYSTEM IMPLEMENTATION

The proposed AI-Based Crop Recommendation System is implemented as an end to end machine learning pipelines that integrates data preprocessing, model inference, and result generation. The system is designed to provide real-time crop recommendations based on user-provided soil and environmental parameters, ensuring practical applicability in agricultural decision-making.

7.1 Input Layer (User Interaction)

The system starts with a screen where the farmer or the agricultural expert types in important details about the soil and the environment around them. This is where the farmer or the agricultural expert puts in information, about the soil and the environment.. These inputs include:

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- Temperature
- Humidity
- Soil pH
- Rainfall

These parameters together show the state of the farm field. The system can take in data, which makes it good for everyday use on farms. It is designed to work in real-life farming situations. The system handles farm conditions. It accepts real-time inputs from the field. This makes it suitable, for farming scenarios.

7.2 Data Preprocessing Layer

Once you give the input parameters the system does some preprocessing. This step makes sure the data works with the trained machine learning model. The preprocessing step is important. It helps the system to get the data ready. The data must be, in a format that machine learning model can understand. The model needs data to be compatible.

The preprocessing steps include:

- Feature Transformation: Input values are converted into a numerical array format suitable for model input.
- Feature Scaling: The input features are standardized using the same scaling technique applied during model training.

This step makes sure that training data and the inference data are consistent. This is important because if they are not the same it can cause problems, with accuracy of the predictions made by the training data and the inference data. Training data and the inference data need to be consistent so that predictions made by the training data and the inference data're correct.

7.3 Model Inference Layer

After we get the input ready we send it to the Random Forest model that we already trained. Random Forest model is made up of decision trees. Each of decision trees looks at the input. Says what kind of crop it thinks it is. Then we use something called majority voting to figure out the answer. We do this by looking at what each decision tree says about the crop category. We choose the answer that most of decision trees agree on. The Random Forest model is really good at this because it uses decision trees to make a prediction, about the crop category.

7.4 Output Layer

The model gives us a number that tells us what type of crop it thinks it is. This number is then used to find the name of the crop. We use a list to match the number with the correct crop name, like a dictionary that works in reverse. The model does this by looking at the number and then finding the crop name so we get the actual name of the crop, like wheat or corn instead of just a number.

Example: 1→Rice

The final result is shown to the user in a way that's easy to understand. This makes the system simple to use for people who're not good, with technology like non-technical users. System is easy to use because the final result is displayed in a readable format.

7.5 Example Prediction

For the given input conditions:

- N = 90
- P = 42
- K = 43
- Temperature $\approx$ 20.87° C
- Humidity $\approx$ 82%
- pH $\approx$ 6.5
- Rainfall $\approx$ 202 mm

The system predicts:

Recommended Crop: Rice

The idea that rice needs a lot of water to grow is something that people who work with farms know is true. Rice usually needs it to be very humid and get a lot of rain. This shows that the computer model that made this prediction is actually pretty good at figuring out how things work in the world with crops, like rice.

7.6 System Workflow

Overall workflow of the system can be summarized as follows:

1. User inputs soil and environmental parameters
2. Input data is preprocessed and standardized
3. Processed data a fed into the trained model
4. Model predicts the most suitable crop
5. Result is displayed to the user

7.7 Practical Applicability

The developed system demonstrates strong real-world usability and can be deployed in various forms, such as:

- Mobile applications for farmers
- Web-based agricultural advisory systems
- Integration with IoT-based smart farming systems

The ability to provide accurate, real-time recommendations makes this system highly valuable for improving agricultural productivity and sustainability.

8. CONCLUSION

This study is about a system that uses Machine Learning(ML) to help farmers choose the right crops. The system looks at things like the nutrients in the soil the weather and how acidic the soil is. It uses a kind of computer program to make sure its predictions are accurate and reliable. The people who made the system tried out a lot of ways to make predictions and found that one way called Random Forest worked the best. This is because it can handle a lot of information understand how different things are related and avoid making mistakes. The system also uses an balanced set of data which helps it make fair predictions for all types of crops. The system is also good because it can make predictions in time based on what the user tells it. This means it could be used on websites, mobile apps and other tools that help farmers make decisions. This could really help farmers choose the crops, which would make their farms more productive and help them use their resources better. The system is also part of an idea called precision agriculture, which is about using technology to make farming better and more sustainable. This study helps bridge the gap between ways of farming and new technology. In short the Artificial Intelligence crop recommendation system is a practical solution for helping farmers choose crops. The study shows that machine learning can really help the farming industry and that using systems is important for sustainable farming. The system uses Artificial Intelligence to make predictions. It looks at things like soil nutrients and the weather. It uses a kind of computer program to make accurate predictions. It can make predictions in time based on user input. It is a part of the precision agriculture concept, which aims to make farming more efficient and sustainable. The Artificial Intelligence crop recommendation system is a tool for farmers. It can help them choose the crops and make their farms more productive. The system is also good for the environment because it helps farmers use their resources better. Overall study suggests that the Artificial Intelligence crop recommendation system is a solution for crop selection. It is effective, scalable and practical. The finding of the study highlight potential of machine learning to transform the farming industry and emphasize the importance of adopting systems, for sustainable farming practices.

9. FUTURE WORK

The proposed AI-Based Crop Recommendation System is really good, at telling us what crops to plant. It works well in real life. The AI-Based Crop Recommendation System can be made better by adding some new features to it. This will help the AI-Based Crop Recommendation System to work for more people and it will be easier to use in the real world.

9.1 Integration with IoT Sensors for Real-Time Data

The current system uses information that people enter by hand like soil nutrients and weather conditions. In future we will be connected the system to Internet of Things (IoT) devices. This way the system can get the data it needs away. The Internet of Things devices can provide real-time soil nutrients and environmental conditions data.

IoT-based sensors can continuously monitor:

- Soil nutrients (N, P, K)
- Soil moisture levels
- Temperature and humidity
- pH levels

This integration would:

- Eliminate manual data entry
- Provide real-time recommendations
- Improve decision-making accuracy
- Enable automated smart farming systems

Such a system can a use in areas where sensors send data to cloud models for crop advice. This can really help farmers make decisions. The sensors can provide data to cloud models. The cloud models then give recommendations on crops. Farmers can use this information to improve their crops. The system is useful, in farming. It uses sensors and cloud models. The goal of this is to helps farmers.

9.2 Use of Deep Learning Models for Improved Accuracy

Although machine learning models, like Random Forest have done a job we can still look into deep learning techniques to make our predictions even more accurate. Machine learning models, Random Forest have shown great results. Deep learning techniques might help us improve prediction accuracy further.

Future work may include:

- Artificial Neural Networks (ANN)
- Deep Neural Networks (DNN)
- Recurrent Neural Networks (RNN) for time-based data

Advantages:

- Ability to capture complex non-linear relationships
- Better performance with large-scale datasets
- Improved adaptability to dynamic agricultural conditions

Additionally, hybrid models combining machine learning models and deep learning models approaches can be investigated.

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